Angular momentum and spin

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physical quantities → operators

$$p_x \longleftrightarrow \hat{p}_x = \frac{\hbar}{i}\frac{\partial}{\partial x}$$

$$x \longleftrightarrow \hat{x} = x$$

Q.: $\vec{L} \longleftrightarrow ?$

$$\vec{L} = \vec{r} \times \vec{p} \rightarrow L_z = x p_y - y p_x$$

assumption (quantum - classical correspondence)

$$\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x$$

or

$$\hat{L}_z = \frac{\hbar}{i}\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right)$$

take wave function $\phi(x,y,z)$

$$\hat{L}_z\phi(x,y,z) = \frac{\hbar}{i}\left(x\frac{\partial\phi}{\partial y} - y\frac{\partial\phi}{\partial x}\right)$$

statement In spherical coordinates

$$\hat{L}_z = \frac{\hbar}{i}\frac{\partial}{\partial\varphi}$$

proof:

$$x = r \sin\vartheta \cos\varphi$$
$$y = r \sin\vartheta \sin\varphi$$
$$z = r \cos\vartheta$$

$$\frac{\partial}{\partial\varphi}\phi(x,y,z) = \frac{\partial\phi}{\partial x}\frac{\partial x}{\partial\varphi} + \frac{\partial\phi}{\partial y}\frac{\partial y}{\partial\varphi} + \frac{\partial\phi}{\partial z}\frac{\partial z}{\partial\varphi} =$$

$\frac{\partial x}{\partial\varphi} = -r\sin\vartheta\sin\varphi = -y$

$\frac{\partial y}{\partial\varphi} = r\sin\vartheta\cos\varphi = x$

$\frac{\partial z}{\partial\varphi} = 0$

$$= -y\frac{\partial\phi}{\partial x} + x\frac{\partial\phi}{\partial y} \quad \checkmark$$

Now $\phi_{nlm}(r,\vartheta,\varphi) = R_{nl}(r)\cdot Y_l^m(\vartheta,\varphi)$

$\sim P_l^m(\vartheta)\cdot e^{i m\varphi}$ — only φ-dep.

$$\hat{L}_z \phi_{nlm} = \frac{\hbar}{i}\frac{\partial}{\partial\varphi}\phi_{nlm} = \frac{\hbar}{i} i m \phi_{nlm}$$

$$\hat{L}_z \phi_{nlm} = \hbar m \cdot \phi_{nlm}$$

(ϕ_{nlm}: eigenstate of $\hat{L}_z$; $\hbar m$: eigenvalue)

$\sim$ $L_z = \hbar m$ | •

Def. $\hat{A}$ some "observable" ($\sim$ operator)

ϕ is an eigenstate of $\hat{A}$ if

$$\hat{A}\phi = a\cdot\phi \qquad a \in \mathbb{C} \sim$$

(a: eigenvalue)

Statement: if e^- is in an eigenstate ϕ_a of $\hat{A}$ of eigenvalue a, $\phi = \phi_a$, then $\hat{A}$ is precisely a, i.e., if we measure $\hat{A}$, then $\hat{A} = a$ with probability 1

Consider, e.g., energy, and assume

$$\hat{H}\,\phi = E_\phi \cdot \phi$$

Then:

$$\langle E \rangle = \langle \hat{H} \rangle = \int d^3\underline{r}\ \phi^*(\underline{r})\ \underbrace{\hat{H}\phi(\underline{r})}_{E_\phi\,\phi(\underline{r})} = \int d^3\underline{r}\ E_\phi\,|\phi(\underline{r})|^2 =$$

$$= E_\phi \underbrace{\int d^3\underline{r}\ |\phi(\underline{r})|^2}_{\parallel\ 1} = E_\phi$$

$$\mathrm{var}(E) = ?$$

$$\langle E^2 \rangle = \langle \hat{H}^2 \rangle = \int d^3\underline{r}\ \phi(\underline{r})\ \underbrace{\hat{H}^2\phi}_{\hat{H}\,\underbrace{\hat{H}\phi}_{E_\phi\cdot\phi}}(\underline{r}) = \int d^3\underline{r}\ |\phi(\underline{r})|^2\ E_\phi^2$$

$$\underbrace{E_\phi^2\,\phi}$$

$$= E_\phi^2$$

$$\Rightarrow \mathrm{var}(E) = \langle E^2 \rangle_\phi - \langle E \rangle_\phi^2 = E_\phi^2 - E_\phi^2 = 0\ !$$

in an eigenstate ϕ_E of $\hat{H}$ the energy is precisely E_ϕ

similarly, in a state ϕ_{nem}

- the energy is precisely $E = -\frac{13,6\ eV}{n^2}$
- L_z is precisely $L_z = \hbar \cdot m$

Q.: $L^2 = ?$ $\quad \hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$

$\hat{L}_x = \frac{\hbar}{i}\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right)$ $\quad \hat{L}_y = \frac{\hbar}{i}\left(z\frac{\partial}{\partial y} - y\frac{\partial}{\partial z}\right)$

Statement: in spherical coordinates:

$$\hat{L}^2 = -\hbar^2\left(\frac{1}{\sin^2\vartheta}\frac{\partial^2}{\partial\varphi^2} + \frac{1}{\sin\vartheta}\frac{\partial}{\partial\vartheta}\sin\vartheta\frac{\partial}{\partial\vartheta}\right)$$

↑ this appeared at separation of variables ↙

$$\frac{1}{\sin^2\vartheta}\frac{\partial^2 Y}{\partial\varphi^2} + \frac{1}{\sin\vartheta}\frac{\partial}{\partial\vartheta}\sin\vartheta\frac{\partial Y}{\partial\vartheta} = -\ell(\ell+1)\,Y$$

$$\Rightarrow \quad \hat{L}^2 Y_\ell^m = \hbar^2 \ell(\ell+1)\, Y_\ell^m$$

or $\quad \hat{L}^2 \phi_{nem} = \hbar^2 \ell(\ell+1)\, \phi_{nem}$

Conclusion: $\phi_{nem} = R_{ne}(r) \cdot Y_e^m(\vartheta, \varphi)$

is a simultaneous eigenstate of

$$H \rightarrow E = -\frac{13{,}6\ eV}{n^2}$$

$$L_z \rightarrow L_z = \hbar \cdot m$$

and $$L^2 \rightarrow L^2 = \hbar^2 \ell(\ell+1)$$

Q: do these states have well-defined L_x?

answer: they cannot

Statement: $\hat{A}$ and $\hat{B}$ observables. Assume we have states $\{\phi\}$ such that they are all simultaneous eigenstates of them

$$\hat{A}\phi = a_\phi \phi \quad , \quad \hat{B}\phi = b_\phi \phi \ .$$

then $$\hat{A}\hat{B}\phi = \hat{A} b_\phi \phi = b_\phi \hat{A}\phi = b_\phi a_\phi \cdot \phi$$

$$= \ldots = \hat{B}\hat{A}\phi \quad \text{for } \forall \phi \ !$$

$\Rightarrow$ $\hat{A}\hat{B} = \hat{B}\hat{A}$ ~ they commute as operators

$\Rightarrow$ $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A} = 0$ ~ their commutator is ZERO

more precisely:

$\hat{A}$ and $\hat{B}$ can have common eigenstates

iff $[\hat{A}, \hat{B}] = 0$!

Heisenberg: $[\hat{p}_x, \hat{x}] = \frac{\hbar}{i} \neq 0$! $\Rightarrow$

cannot have simultaneous eigenstates

$\sim$ p_x and x cannot have a precise value simultaneously ...

What about L_x, L_y, L_z ?

$[\hat{L}_z, \hat{L}_x] = i\hbar \hat{L}_y$; $[\hat{L}_x, \hat{L}_y] = i\hbar \hat{L}_z$

$[\hat{L}_y, \hat{L}_z] = i\hbar \hat{L}_x$ ← homework !

they cannot have simultaneous eigenstates,

(unless $\ell = 0$)

However

$[\hat{L}^2, \hat{L}_z] = 0$ and $[\hat{H}, \hat{L}^2] = [\hat{H}, \hat{L}_z] = 0$

L_z, L^2 and H can have

simultaneous eigenstates.

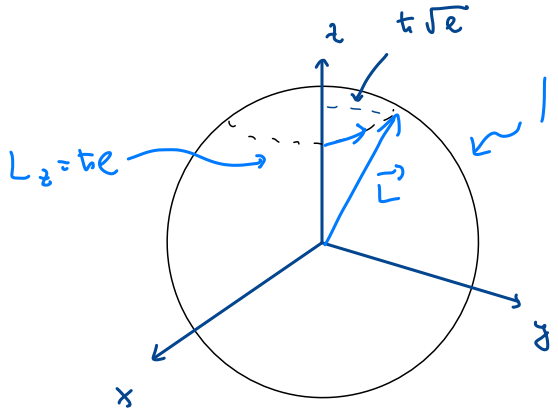
Consider $L^2 = \hbar^2 \ell(\ell+1)$

then $\hat{L}_z = -\hbar\ell, \dots, \hbar\ell$

($\hbar\ell$: maximal value)

but $L_z^{max} = \hbar \cdot \ell < |\vec{L}| = \hbar\sqrt{\ell(\ell+1)}$

semi-classical picture:



$\vec{L}$ rotates around $\hat{z}$, but cannot be perfectly aligned...

Q.: $\langle L_x^2 \rangle = ?$ if $L_z = \hbar\ell$

A: $L^2 = L_x^2 + L_y^2 + L_z^2$

$\Rightarrow \langle L^2 \rangle = \langle L_x^2 \rangle + \langle L_y^2 \rangle + \langle L_z^2 \rangle$

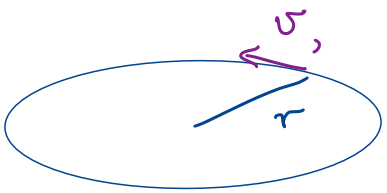
$\langle L^2 \rangle$: $\hbar^2 \ell(\ell+1)$; $\langle L_z^2 \rangle$: $\hbar^2 m^2 \to \hbar^2 \ell^2$

$$\langle L_x^2 \rangle = \langle L_y^2 \rangle = \frac{1}{2}\hbar^2 \cdot \ell \quad \Rightarrow \quad \Delta L_x = \Delta L_y = \hbar\sqrt{\frac{\ell}{2}}$$

Can we measure m ?

m ~ magnetic quantum number !

circular motion



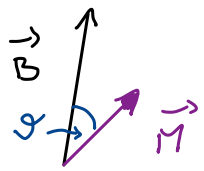
magnetic moment

$$M = A \cdot I = r^2 \pi \cdot \frac{v}{2\pi \cdot r} \cdot (-e)$$

$$= -\frac{e}{2m_e} \cdot \underbrace{(m_e v \cdot r)}_{L}$$

$$\vec{M} = -\frac{e}{2m_e} \cdot \vec{L}$$

magnetic moment couples to external field;



$$\Delta E = -B \cdot M \cdot \cos(\vartheta) = -\vec{B} \cdot \vec{M}$$

consider $\vec{B} \parallel z$, then

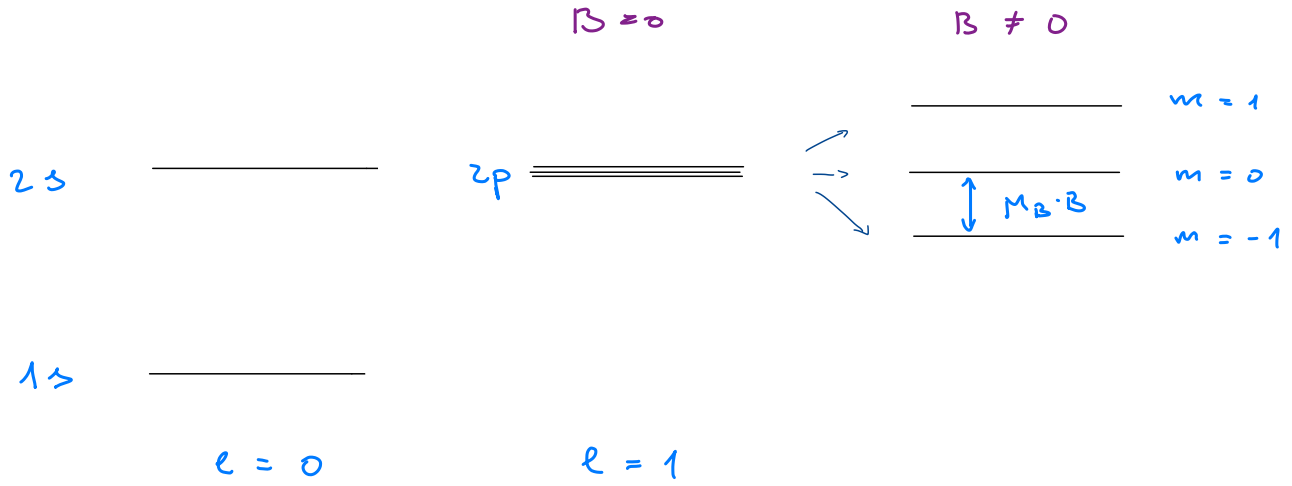
$$\Delta E = -B \cdot M^z = B \cdot \frac{e}{2m_e} \underbrace{L_z}_{\hbar \cdot m} =$$

$$\Delta E = B \cdot \frac{e\hbar}{2m_e} \cdot m$$

μ_B Bohr magneton

Zeeman splitting

Splitting of $\ell \neq 0$ levels in $B \neq 0$



size of splitting? $B = 1\,T \Rightarrow B\mu_B \sim 0{,}5 \cdot 10^{-4}\,eV$

very small

but can be used for quantum technology ...

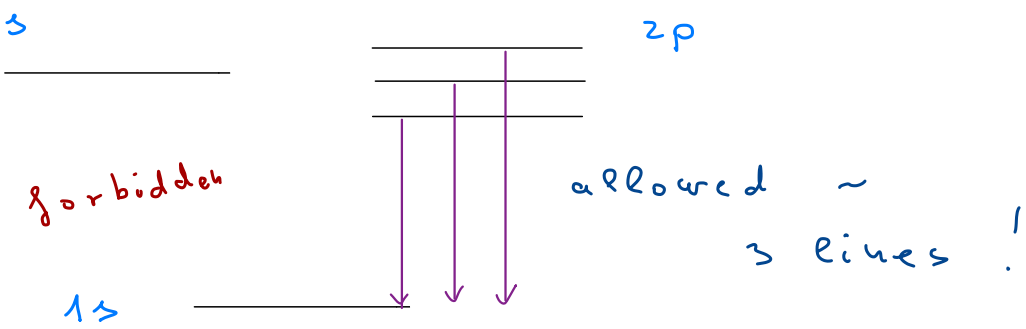
Observation in atomic spectra ?

"Allowed" transitions:

$$\ell \rightarrow \ell \pm 1$$

$$m \rightarrow m \pm 1,\ m$$

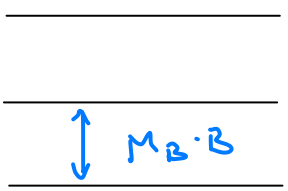
$$n \rightarrow \text{any } n$$



Stern – Gerlach experiment

use inhomogeneous field

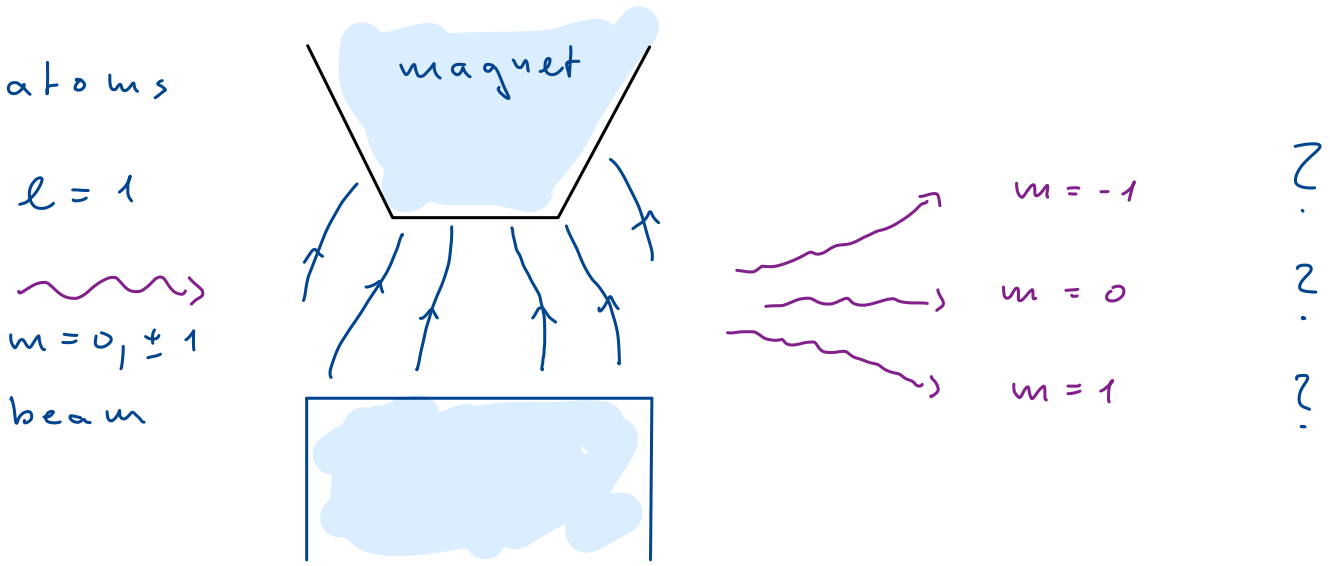
$\vec{B} \rightarrow \vec{B}(\underline{r})$



← low field seeker

indifferent

← high field seeker



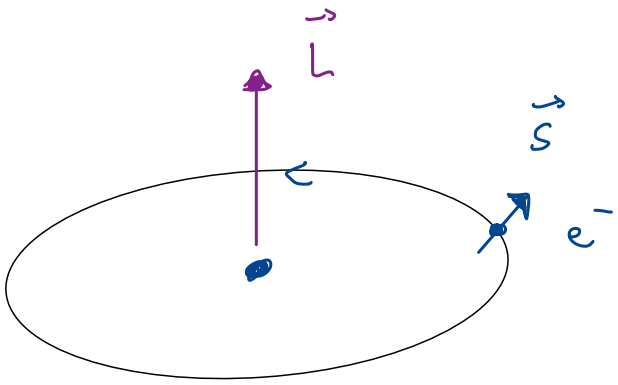
experiment with Ag ⇒ beam split in TWO!

ℓ must be $\frac{1}{2}$! $m = \pm \frac{1}{2}$

explanation: e^- spins eternally! ~ planet!

$L \rightarrow$ orbital angular momentum

$S \rightarrow$ its own electron's own angular momentum



experiment $\Rightarrow$ $S^z = \hbar m_s$, $m_s = \pm \frac{1}{2}$ $\Rightarrow$ $s = \frac{1}{2}$

$S^2 \overset{?}{=} \hbar^2 s(s+1) = \hbar^2 \frac{3}{4}$?

Pauli matrices: ~ spin operators

internal angular momentum:

operator? $\hat{S}_z$ $\rightarrow$ eigenvalues $\pm \frac{\hbar}{2}$

~ matrix $\Rightarrow$ $\hat{S}_z = \frac{\hbar}{2} \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$

σ_z Pauli matrix

eigenstates:

$m_s = \frac{1}{2}$ $\Rightarrow$ $\chi_\uparrow \equiv \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ state ~ "spin up" $\uparrow$

$m_s = -\frac{1}{2}$ $\Rightarrow$ $\chi_\downarrow \equiv \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ ~ "spin down" $\downarrow$

e^- state is characterized by

$$n, \ell, m_\ell, m_s$$

$$\sim m_s$$

$$\Rightarrow \{n, \ell, m, \uparrow\} \Rightarrow \phi_{nem}(r,\vartheta,\varphi)\cdot\chi_\uparrow$$

$$\{n, \ell, m, \downarrow\} \Rightarrow \phi_{nem}(r,\vartheta,\varphi)\cdot\chi_\downarrow$$

Q: what are $\hat{S}_x$ and $\hat{S}_y$?

assume ~ angular momentum !

$$[\hat{S}_x, \hat{S}_y] = i\hbar \hat{S}_z \quad + \text{cyclic relations}$$

$$\Rightarrow \hat{S}_x = \frac{\hbar}{2}\underbrace{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}}_{\sigma_x}, \quad \hat{S}_y = \frac{\hbar}{2}\underbrace{\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}}_{\sigma_y}$$

Nice properties: $\sigma_z^2 = \sigma_x^2 = \sigma_y^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$$\Rightarrow \hat{S}^2 = \hat{S}_x^2 + \hat{S}_y^2 + \hat{S}_z^2 = \hbar^2 \frac{3}{4}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$s(s+1), \quad s = \frac{1}{2}$$

homework: verify

$$[\hat{S}_x, \hat{S}_y] = i\hbar \hat{S}_z$$

$\hat{S}_x$, $\hat{S}_y$ have also eigenvalues $\pm \frac{\hbar}{2}$

$S_x = \frac{\hbar}{2}$ state, e.g.

$$\hat{S}_x \cdot \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = + \frac{\hbar}{2} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$\Rightarrow \quad \frac{\hbar}{2} \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = 0 \quad \Rightarrow \quad \begin{pmatrix} \alpha_+ \\ \beta_+ \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} =$$

~ mixture of "up" and "down"

similarly $\hat{S}_x \cdot \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = - \frac{\hbar}{2} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \Rightarrow$

$$\begin{pmatrix} \alpha_- \\ \beta_- \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} (\chi_\uparrow - \chi_\downarrow)$$

most particles have a spin!

neutron, proton neutrino, positron	$\frac{1}{2}$
photon, gluon, $W_\pm$, Z	1
pion	0
nuclei	$0, \frac{1}{2}, 1, \frac{3}{2}, \ldots$